

ENTHUSIAST ADVANCE COURSE

PHASE : MEA & MEPS-I (A)

TARGET : PRE-MEDICAL 2024

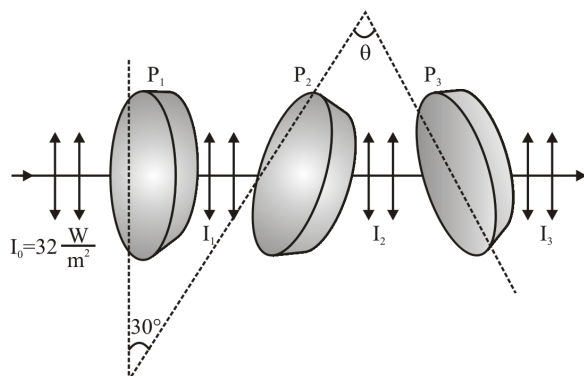
Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 21-05-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	3	2	4	2	3	3	1	1	3	4	1	4	3	2	2	1	2	3	3	4	1	4	3	4	1	2	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	1	1	3	4	3	3	4	2	4	1	3	2	2	1	2	1	3	2	4	4	3	3	1	2	4	2	4	2	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	3	2	2	1	2	3	2	2	2	4	2	1	3	1	2	1	3	2	3	3	1	1	2	3	2	1	4	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	1	1	3	3	4	3	2	2	4	3	2	1	2	3	3	3	2	3	2	2	4	1	4	4	2	2	3	1	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	3	2	3	1	4	4	1	4	4	4	3	3	4	1	2	1	3	2	1	2	4	3	4	1	2	4	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	1	3	4	2	1	1	1	2	3	1	2	2	2	2	3	2	2	4	2	3	2	1	1	1	2	2	3	1	1
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	3	2	2	3	4	1	3	2	4	4	1	3	4	3	4	2	1	4	3										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**1. **Ans (2)**Angle between P_1 and $P_2 = 30^\circ$ (given)Angle between P_2 and $P_3 = \theta = 90^\circ - 30^\circ = 60^\circ$ The intensity of light transmitted by P_1 is

$$I_1 = \frac{I_0}{2} = \frac{32}{2} = 16 \frac{W}{m^2}$$

According to Malus law the intensity of light transmitted by P_2 is

$$I_2 = I_1 \cos^2 30^\circ = 16 \left(\frac{\sqrt{3}}{2} \right)^2 = 12 \frac{W}{m^2}$$

Similarly intensity of light transmitted by P_3 is

$$I_3 = I_2 \cos^2 \theta = 12 \cos^2 60^\circ = 12 \left(\frac{1}{2} \right)^2 = 3 \frac{W}{m^2}$$

2. **Ans (2)**

Brewster's law

3. **Ans (3)**From Brewster's law $\mu = \tan i_p$

$$\Rightarrow \frac{c}{v} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow v = \frac{c}{\sqrt{3}} = \frac{3 \times 10^8}{\sqrt{3}} = \sqrt{3} \times 10^8 \text{ m/sec}$$

4. **Ans (2)**

$$\frac{h_c}{h_o} = \frac{f}{f-u}$$

$$\frac{-1}{5} = \frac{-10}{-10-u}$$

$$10+u = -50$$

$$u = -60 \text{ cm}$$

5. **Ans (4)**

The two slabs will shift the image a distance

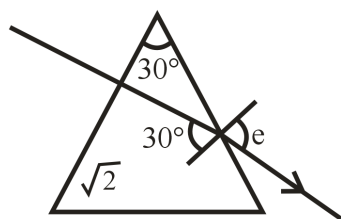
$$d = 2 \left(1 - \frac{1}{\mu} \right) t = 2 \left(1 - \frac{1}{1.5} \right) (1.5) = 1.0 \text{ cm}$$

Therefore, final image will be 1 cm above point P.

6. **Ans (2)**

$$r = \frac{h}{\sqrt{\mu^2 - 1}} = \frac{4}{4} \times 3 = 3 \text{ m}$$

Diameter = 6 m

7. **Ans (3)**

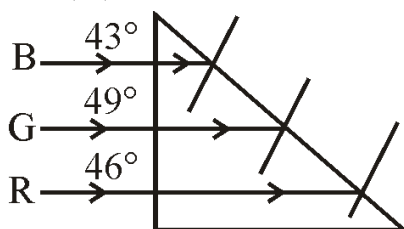
$$\sqrt{2} \sin 30^\circ = 1 \sin e$$

$$\boxed{e = 45^\circ}$$

$$\delta = i + e - A$$

$$= 0 + 45^\circ - 30^\circ$$

$$\boxed{\delta = 15^\circ}$$

8. **Ans (3)**For each ray angle of incidence = 45° For blue and green $i > \theta_c$

so B and G will get reflected while red will get refracted

9. **Ans (1)**

$$\omega = \frac{\mu_V - \mu_R}{\mu_Y - 1} = \frac{1.62 - 1.42}{1.5 - 1} = 0.4$$

10. **Ans (1)**

$$\frac{f_l}{f_a} = \frac{a\mu_g - 1}{i\mu_g - 1} = \frac{1.5 - 1}{\frac{1.5}{1.75} - 1} = \frac{1.75 \times 0.50}{0.25} = -3.5$$

$$\therefore f_l = -3.5 f_a \Rightarrow f_l = +3.5 R (\because f_a = R)$$

Hence on immersing the lens in the liquid, it behaves as a converging lens of focal length $3.5R$.11. **Ans (3)**

$$u = -25 \text{ cm}$$

$$v = -60 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= -\frac{1}{60} + \frac{1}{25} = \frac{7}{300}$$

$$P = \frac{7}{300} \times 100 = +2.33D$$

12. **Ans (4)**

$$m_0 = \frac{f_0}{f_0 + u} = \frac{4}{4 - 5} = -4$$

$$m_e = \frac{D}{f_e} + 1 = \frac{25}{10} + 1 = 3.5$$

$$\therefore MP = m_0 \times m_e = -4 \times 3.5 = -14$$

13. **Ans (1)**

At extreme position, velocity & hence momentum becomes zero

14. **Ans (4)**

$$x = \sin \left(\frac{2\pi}{8} t \right)$$

$$a = -\omega^2 x = -\frac{2\pi}{8} \times \frac{2\pi}{8} \times \sin \left(\frac{2\pi}{8} \times \frac{4}{3} \right)$$

$$= \frac{\pi^2}{16} \times \frac{\sqrt{3}}{2} = \frac{-\sqrt{3}\pi^2}{32}$$

15. **Ans (3)**

$$y = a \sin \omega t$$

$$\text{For 1 sec : } y_1 = a \sin \omega$$

$$\text{For 2 sec : } y_2 = a \sin 2\omega$$

$$\frac{y_1}{y_2} = \frac{\sin \omega}{\sin 2\omega} = \frac{1}{2 \cos \omega} = \frac{1}{2 \cos \frac{2\pi}{T}}$$

$$= \frac{1}{2 \cos \frac{2\pi}{8}} = \frac{1}{2 \left(\frac{1}{\sqrt{2}} \right)} = \frac{1}{\sqrt{2}}$$

$$y_2 = \sqrt{2} y_1$$

$$\text{Distance covered in 2nd second} = (\sqrt{2} - 1) y_1$$

$$\text{Required ratio} = \frac{1}{(\sqrt{2} - 1)}$$

16. Ans (2)

A body oscillates simple harmonically with a time period of 2s

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi \text{ rad/sec}$$

The velocity of a body is given by :

$$u = a\omega \cos \omega t \quad (a = \text{amplitude})$$

Kinetic energy of a body is,

$$KE = \frac{1}{2}mu^2 = \frac{1}{2}ma^2\omega^2\cos^2\omega t$$

Total energy of a body is,

$$E = \frac{1}{2}mu_{\max}^2 = \frac{1}{2}ma^2\omega^2$$

As per question :

$$KE = \frac{75}{100}E$$

$$\therefore \frac{1}{2}ma^2\omega^2\cos^2\omega t = \frac{3}{4} \times \frac{1}{2}ma^2\omega^2$$

$$\text{or } \cos^2\omega t = \frac{3}{4}$$

$$\text{or } \cos\omega t = \frac{\sqrt{3}}{2}$$

$$\text{or } \omega t = \frac{\pi}{6}$$

$$\therefore t = \frac{\pi}{6\omega} = \frac{\pi}{6\pi} = \frac{1}{6} \text{ sec}$$

17. Ans (2)

$$x = A\sin(\omega t + \pi/4)$$

speed will be zero at $x = +A$ first time

$$A = A\sin(\omega t + \pi/4)$$

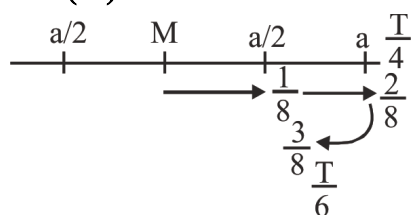
$$\sin\left(\omega t + \frac{\pi}{4}\right) = 1$$

$$\Rightarrow \omega t + \frac{\pi}{4} = \frac{\pi}{2}$$

$$\omega t = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$\frac{2\pi}{T}t = \frac{\pi}{4}$$

$$t = \frac{T}{8}$$

18. Ans (1)

$$\text{Total time} = \frac{T}{4} + \frac{T}{6} = \frac{5T}{12}$$

19. Ans (2)

$$\begin{aligned} V &= \omega \sqrt{A^2 - (A/2)^2} \\ &= \frac{2\pi}{T} \times \sqrt{3} \frac{A}{2} \\ &= \sqrt{3} \frac{\pi A}{T} \end{aligned}$$

20. Ans (3)

$$\begin{aligned} K &= \frac{1}{n} \text{ up} \Rightarrow \frac{1}{2} m\omega^2(A^2 - x^2) = \frac{1}{n} \frac{1}{2} m\omega^2 x^2 \\ \text{that } A^2 - x^2 &= \frac{x^2}{n} \quad \text{or } x^2 \left(1 - \frac{1}{n}\right) = A^2 \\ x &= \sqrt{\frac{n}{n+1}} A \end{aligned}$$

21. Ans (3)

$$\begin{aligned} k_{\text{eq}} &= \frac{k}{2} & k_{\text{eq}} &= \frac{2k}{3} \\ n_1 &= \frac{1}{2\pi} \sqrt{\frac{k/2}{m}} & n_2 &= \frac{1}{2\pi} \sqrt{\frac{2k/3}{m}} \\ \text{So, } \frac{n_1}{n_2} &= \frac{\sqrt{3}}{2} \end{aligned}$$

22. Ans (4)

In damped oscillation, amplitude goes on decaying exponentially.

$$a = a_0 e^{-bt} \text{ where } b = \text{damping coefficient.}$$

Initially, $\frac{a_0}{3} = a_0 e^{-b \times 100T}$, T = time of one oscillation

$$\text{or } \frac{1}{3} = e^{-100bT} \quad \dots(i)$$

$$\text{Finally, } a = a_0 e^{-b \times 200T} \text{ or } a = a_0 [e^{-100bT}]^2$$

$$\text{or } a = a_0 \times \left[\frac{1}{3}\right]^2 \quad [\text{From (i)}]$$

$$\text{or } a = a_0/9$$

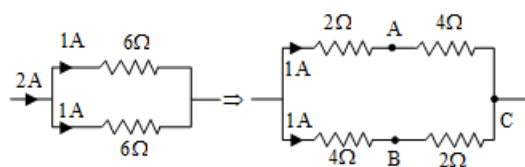
25. Ans (3)

$$\begin{aligned} R_{AB} &= \frac{5}{6}R \parallel \frac{5}{6}R \\ &= \frac{5}{12}R \end{aligned}$$

26. Ans (4)

$$\frac{12}{X+6} = \frac{1/2}{0.5} \Rightarrow X+6 = 12 \Rightarrow X = 6\Omega$$

28. Ans (2)



$$V_A - V_C = 1 \times 4 = 4V \quad \dots (1)$$

$$V_B - V_C = 1 \times 2 = 2V \quad \dots (2)$$

$$V_A - V_B = 4 - 2 = 2V$$

SECTION-B

36. Ans (3)

Laws of reflection are valid for all surfaces.

37. Ans (3)

$$d_{app} = \frac{d_1}{n_1} + \frac{d_2}{n_2}$$

38. Ans (4)

$$\theta_c = \sin^{-1} \left(\frac{1.8 \times 10^8}{2.4 \times 10^8} \right)$$

$$\theta_c = \sin^{-1} \left(\frac{3}{4} \right)$$

39. Ans (2)

$$\mu = \frac{\sin \left(\frac{A + \delta m}{2} \right)}{\sin A/2} = \frac{\sin A}{\sin A/2} = 2 \cos A/2$$

$$\text{for } A = 90, \mu = \sqrt{2} \quad \text{for } A = 0, \mu = 2$$

$$\therefore \sqrt{2} < \mu < 2$$

40. Ans (4)

Given, concave lens, $f = -21$ cm, $u = -14$ cmObject size $O = 3$ cm

Applying lens formula,

$$v = \frac{uf}{u+f} = \frac{(-14) \times (-21)}{(-14) + (-21)} = \frac{-45}{5} \text{ cm} = -8.4 \text{ cm}$$

Using the formula of magnification of lens

$$m = \frac{v}{u} = \frac{I}{O}, \text{ where } I = \text{height of image}$$

$$\Rightarrow \frac{(-8.4)}{-1.4} = \frac{I}{+3} \Rightarrow I = 1.8 \text{ cm}$$

Magnification of a lens,

$$M = \frac{v}{u} = \frac{-8.4}{-1.4} = 0.6$$

41. Ans (1)

$$M.P. = \frac{f_0}{f_c} = 14$$

$$f_0 = 14f_c$$

$$= 14 \times 5$$

$$f_0 = 70 \text{ cm}$$

$$\text{Length} = f_0 + f_c \Rightarrow 70 + 5$$

$$\Rightarrow 75 \text{ cm}$$

42. Ans (3)

$$y = a[\sin \omega t + \cos \omega t]$$

$$y = a\sqrt{2} \left(\frac{1}{\sqrt{2}} \sin \omega t + \frac{1}{\sqrt{2}} \cos \omega t \right)$$

$$y = a\sqrt{2} \sin \left(\omega t + \frac{\pi}{4} \right)$$

43. Ans (2)

$$A\omega^2 = 5\pi^2 \quad (3\pi)^2 = \omega^2 (A^2 - 16)$$

$$\frac{5}{9} = \frac{A}{A^2 - 16}$$

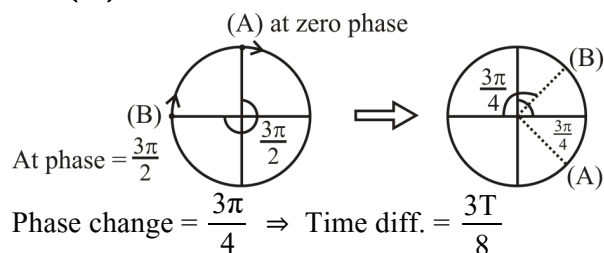
$$5A^2 - 80 - 9A = 0$$

$$A = 5 \text{ cm}$$

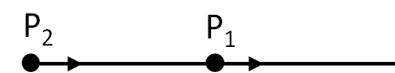
$$\omega = \pi$$

$$T = 2 \text{ sec.}$$

44. Ans (2)



OR



$$\Delta\phi = 270^\circ$$

$$\phi_1 = 135^\circ = \frac{3\pi}{4}$$

$$t = \frac{3T}{8}$$

45. Ans (1)

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T' = 2\pi\sqrt{\frac{1.21\ell}{g}} = 1.1 \left(2\pi\sqrt{\frac{\ell}{g}} \right) = 1.1 T$$

$$\Rightarrow \% \text{ increase in time period will be } 10\%$$

SUBJECT : CHEMISTRY**SECTION-A**

52. **Ans (3)**
 NH_4^+ ion does not have any lone pair, so not used as ligand.
54. **Ans (1)**
 NO is $3e^-$ donor
 $\text{EAN(Fe)} = 26 - 0 + 2 \times 2 + 3 \times 2$
 $= 36$ (CO) NO
55. **Ans (2)**
 Non-ionisable secondary valencies have directional properties.
56. **Ans (4)**
 due to presence of ambidentate ligand CN/NC
63. **Ans (3)**
 $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ \rightarrow$ is the parameter for crystal triclinic.
64. **Ans (2)**
 $\frac{r^+}{r^-} = \frac{95}{181} = 0.52$ (this value is present in between normal range so, C.N. remain 6)
65. **Ans (2)**
 The empty space in figure is octahedral void because it is surrounded by six spheres.
76. **Ans (1)**
 $k = \frac{2.303}{t} \log \frac{C_0}{C_t}$
 $\frac{t_{0.999}}{t_{0.5}} = \frac{\frac{2.303}{k} \log \frac{100}{0.1}}{\frac{2.303}{k} \log \frac{100}{50}} = \frac{3}{0.3} = \frac{10}{1} = 10$

SECTION-B

87. **Ans (2)**
 $[\text{PtCl}_6]^{-2}$
 $\downarrow$
 at no. = 78, O. S. (Pt) = +4
 $78 - 4 + 2 \times 6$
 $74 + 12 = 86$

- 92.
- Ans (1)**

In bcc $\sqrt{3}a = 4r$

$$\frac{\sqrt{3}}{2} a = 2r$$

$$\therefore 2r = \frac{\sqrt{3}}{2} \times 4 \times 10^{-10} = 3.46 \times 10^{-10} \text{ m}$$

- 93.
- Ans (1)**

Total No. of octahedral void per atom in

$$\text{ccp} = \frac{4}{4} = 1$$

- 97.
- Ans (3)**

$$[A]_t = [A] - Kt = 1 - 0.001 \times 10 \times 60$$

$$[A]_t = 0.4 \text{ M}$$

$$[B]_t = 0.001 \times 10 \times 60 = 0.6 \text{ M}$$

SUBJECT : BIOLOGY-I**SECTION-A**

102. **Ans (2)**
 NCERT-XII Pg. # 233
103. **Ans (1)**
 NCERT-XI Pg. # 236
104. **Ans (2)**
 NCERT-XII Pg. # 242
105. **Ans (3)**
 NCERT-XII Pg. # 235
107. **Ans (3)**
 NCERT-XII, Pg. # 243, 244
108. **Ans (2)**
 NCERT XIIth, Pg. # 242, 3rd para
109. **Ans (3)**
 NCERT-XII, Pg. # 243
110. **Ans (2)**
 NCERT-XII, Pg. # 224
113. **Ans (1)**
 NCERT XII, Pg. # 254
114. **Ans (4)**
 NCERT Pg. # 245
116. **Ans (2)**
 NCERT-XII, Pg. # 254

117. **Ans (2)**
NCERT, Pg. # 253

119. **Ans (1)**
NCERT-XII, Pg. 246

120. **Ans (3)**
NCERT-XII, Pg. # 226

121. **Ans (4)**
NCERT-XII, Pg. # 226

122. **Ans (4)**
NCERT Pg. # 222

123. **Ans (3)**
NCERT-XII, Pg. 226

124. **Ans (2)**
NCERT-XII, Pg. # 226
statements a, c

125. **Ans (3)**
NCERT-XII, Pg. # 248

SECTION-B

136. **Ans (1)**
NCERT-XII, Pg. # 245

137. **Ans (2)**
NCERT-XII, Pg. # 243

139. **Ans (3)**
NCERT-XII, Pg. # 254

140. **Ans (2)**
NCERT-XII, Pg. # 254

141. **Ans (1)**
NCERT-XII, Pg. # 249

143. **Ans (4)**
NCERT-XII, Pg. # 225

144. **Ans (3)**
NCERT-XII, Pg. # 223 (Fig. 13.3)

145. **Ans (4)**
NCERT-XII, Pg. # 220

SUBJECT : BIOLOGY-II

SECTION-A

153. **Ans (3)**
NCERT, Pg#44, Para.-3.1

163. **Ans (2)**
NCERT Pg. # 110, Para 6.5.3

168. **Ans (2)**
NCERT Pg # 90

171. **Ans (3)**
NCERT XII, Pg # 122, 6.10

173. **Ans (1)**
NCERT-XII Pg. No. # 115 Para- 6.8

177. **Ans (2)**
NCERT (XII) Pg. # 110,111

179. **Ans (1)**
NCERT Pg. # 00

180. **Ans (1)**
NCERT-XII (E) Pg#88, PARA-5.7

181. **Ans (1)**
NCERT Pg. # 114

182. **Ans (3)**
NCERT XII Pg.# 109

185. **Ans (3)**
NCERT-XII Pg. # 99,111

SECTION-B

193. **Ans (3)**
NCERT (XII) Pg. # 115 para 3

195. **Ans (3)**
NCERT XII, Pg. # 111

197. **Ans (2)**
NCERT XII, Pg. # 117, 6.8.1

198. **Ans (1)**
NCERT-XII, Pg. # 121